In some implementations the encoder neural network includes 6 encoder subnetworks, each having 8 attention heads. The system processes a batch of 25 sequences in parallel and returns the output within 10 milliseconds. Other implementations use 20 layers or more, depending on the available memory of the device. In some implementations the encoder neural network includes 6 encoder subnetworks, each having 8 attention heads. The system processes a batch of 25 sequences in parallel and returns the output within 10 milliseconds. Other implementations use 20 layers or more, depending on the available memory of the device.

The decoder generates the output sequence one position at a time. In one example the vocabulary contains 30 thousand tokens and the beam width is 5 candidates. Training proceeds for 15 epochs with a learning rate that decays after 40 steps, and the final model is evaluated on 55 held out documents. The decoder generates the output sequence one position at a time. In one example the vocabulary contains 30 thousand tokens and the beam width is 5 candidates. Training proceeds for 15 epochs with a learning rate that decays after 40 steps, and the final model is evaluated on 55 held out documents.